

George Emil Sadek

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Professional Summary

Final-year Computer Engineering student with hands-on experience building LLM-enabled applications, retrieval-augmented generation (RAG) systems, backend APIs, and Dockerized development workflows. Skilled in Python, FastAPI, Flask, LangChain, LlamaIndex, CrewAI, SQL, and Linux, with strong interest in agentic systems, AI-assisted engineering workflows, and production-oriented backend development. Experienced in prompt design, retrieval pipeline construction, API integration, debugging complex systems, and communicating technical work clearly in English.

Technical Skills

Programming: Python, JavaScript, Java, C, C++, SQL

LLM & Agentic Systems: RAG, LangChain, LlamaIndex, CrewAI, prompt engineering, embeddings, vector retrieval, LLM evaluation, AI-assisted workflows

Backend & APIs: FastAPI, Flask, REST APIs, backend service design, third-party API integration, JSON, request handling

Databases & Data: MySQL, SQLite, SQL Server, Pandas, NumPy

Dev Tools & Deployment: Docker, Git, Linux, VS Code, Jupyter, Google Colab

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, NLP, deep learning, computer vision

Selected Projects

Student Helper – RAG-Based Learning Assistant

Personal Project

- Built a retrieval-augmented assistant that answers questions using user-provided study materials instead of generic responses.
- Designed the end-to-end pipeline for document ingestion, chunking, embeddings, semantic retrieval, and grounded response generation.
- Developed backend logic for query handling and response orchestration, improving answer relevance and reducing hallucinated output.
- Applied prompt engineering and evaluation practices to improve context selection, answer quality, and reliability.

Rehab Helper – Medical AI Assistant and Web Platform

Graduation Project

- Contributing to a scalable platform that combines medical detection models with an AI assistant for patient support and guidance.
- Built a knowledge-grounded assistant to answer patient questions within safe boundaries and direct sensitive cases toward therapist follow-up.
- Worked on backend-oriented application design, AI workflow integration, and system structure for maintainable feature growth.
- Helped shape a product flow that includes therapist feedback, messaging, and patient progress support around the AI experience.

Agentic Workflow Prototyping

Applied AI Engineering

- Prototyped agent-based workflows using LLM orchestration tools such as CrewAI, LangChain, and LlamaIndex.
- Explored tool-using agents for structured actions including search, task routing, external data access, and workflow automation.
- Practiced debugging and refining AI-generated outputs through better constraints, prompt design, and system decomposition.

Email Spam Detection

Machine Learning Project

- Developed an NLP-based email classification system using Python and scikit-learn with TF-IDF-based text features.
- Built a clean preprocessing and model evaluation pipeline, achieving strong classification accuracy on benchmark data.

- Strengthened practical understanding of text preprocessing, feature engineering, error analysis, and model debugging.

Education

B.Sc. in Computer Engineering

Sep 2021 – Jun 2026

Faculty of Engineering at Shoubra, Benha University

CGPA: 3.4/4.0

Leadership, Activities & Awards

AI Leader, IEEE Shoubra Team

- Supported AI-focused learning activities and helped peers with machine learning and NLP topics.

Shoubra Racing Team – Shell Eco-marathon Autonomous Programming Competition

- Contributed to autonomous systems work involving perception, simulation, and problem-solving in engineering environments.
- Achieved **2nd Place** with the team in the Shell Eco-marathon Autonomous Programming Competition.

Training & Certifications

AI Training – NTI | Data Analysis Training – NTI | Entrepreneurship Training – ITIDA / Plan International
Machine Learning Specialization (Stanford / Andrew Ng), Deep Learning Specialization, AWS ML Foundations,
HTML/CSS/JavaScript coursework

Additional Information

Languages: Arabic (Native), English (Professional)

Availability: Open to full-time, on-site software engineering roles in Cairo